

Incident Detection

6.pcap

How many events were generated during the attack?

Let's run the analysis:

```
(kali@kali) ~/suricata_lab/lab1
$ suricata -f 6.pcap
i: suricata: This is Suricata version 8.0.3 RELEASE running in USER mode
i: mpm-hu: Failed to create Hyperscan cache folder, make sure the parent folder is writeable or adjust sgh-mpm-caching-path setting (/var/lib/suricata/cache/sgh)
i: unix-manager: Unix socket: UNIX socket bind(/var/run/suricata/suricata-command.socket) error: Address already in use
i: unix-manager: Unable to create unix command socket
i: threads: Threads created -> RX: 1 W: 4 FM: 1 FR: 1 Engine started.
i: suricata: Signal Received. Stopping engine.
i: pcap: read 1 file, 5880 packets, 462888 bytes
```

Let's write a script that counts the total number of events, determines the distribution of events by type (alert, http, dns, flow, etc.), displays the most frequent IDS trigger signatures, analyzes TLS Server Name Indication (SNI), identifies HTTP hosts, displays DNS queries, and outputs downloaded files:

```
#!/bin/bash
filename="./eve.json"
if [ ! -f "$filename" ]; then
    echo "eve.json not found at $filename"
    exit 1
fi
echo "Total number of events:"
jq -c '.' "$filename" | wc -l
echo "Event types:"
jq -r '.event_type' "$filename" | sort | uniq -c | sort -nr
echo "Alert signatures:"
jq -r 'select(.event_type="alert") | .alert.signature' "$filename" \
| sort | uniq -c | sort -nr
echo "Signature IDs:"
jq -r 'select(.event_type="alert") | .alert.signature_id' "$filename" \
| sort | uniq -c | sort -nr
echo "TLS Server Names (SNI):"
jq -r 'select(.event_type="tls") | .tls.sni // empty' "$filename" \
| grep -v '^$' | sort | uniq -c | sort -nr
echo "HTTP Hosts:"
jq -r 'select(.event_type="http") | .http.hostname // empty' "$filename" \
| grep -v '^$' | sort | uniq -c | sort -nr
echo "DNS Queries:"
jq -r 'select(.event_type="dns") | .dns.query // empty' "$filename" \
| grep -v '^$' | sort | uniq -c | sort -nr
echo "Observed Filenames/URLs:"
jq -r 'select(.event_type="fileinfo") | .http.url // .fileinfo.filename // empty' "$filename" \
| grep -v '^$' | sort | uniq -c | sort -nr
```

After analyzing Suricata logs, the following was established: total number of events: 579, number of IDS alerts: 19. Among the recorded events are: HTTP requests, DNS requests, TLS connections, network flows. The main internal host is 192.168.1.216. This host actively interacted with the external IP address 2.56.57.108. HTTP requests, TLS connections, and executable file downloads were also recorded, which are signs of infected system activity.

```
(kali@kali) ~/suricata_lab/lab1
$ ./suricata_sum.sh
Total number of events:
579
Event types:
167 flow
88 dns
87 smb
77 dcerpc
48 krb5
33 ldap
24 tls
19 alert
16 http
14 fileinfo
2 mdns
2 anomaly
1 stats
1 dhcp
```

```

Alert signatures:
9 ET MALWARE Vidar/Arkei/Megumin/Oski Stealer HTTP POST Pattern
3 ET HUNTING SUSPICIOUS Dotted Quad Host MZ Response
2 SURICATA Applayer Detect protocol only one direction
1 ET INFO PE EXE or DLL Windows file download HTTP
1 ET INFO NTLMv1 Session Setup Response - Challenge
1 ET INFO NTLM Session Setup Request - Negotiate
1 ET INFO NTLM Session Setup Request - Auth
1 ET HUNTING Suspicious Zipped Filename in Outbound POST Request (Chrome_Default.txt)

Signature IDs:
9 2034813
3 2021076
2 2260002
1 2067087
1 2067086
1 2067085
1 2033886
1 2018959

TLS Server Names (SNI):
6 v10.events.data.microsoft.com
4 settings-win.data.microsoft.com
3 www.bing.com
2 self.events.data.microsoft.com
1 v20.events.data.microsoft.com
1 pti.store.microsoft.com
1 kv801.prod.do.dsp.mp.microsoft.com
1 geo.prod.do.dsp.mp.microsoft.com
1 fe3cr.delivery.mp.microsoft.com
1 fe2cr.update.microsoft.com
1 cp801.prod.do.dsp.mp.microsoft.com
1 client.wns.windows.com
1 api.msn.com

HTTP Hosts:
9 2.56.57.108
3 download.windowsupdate.com
3 au.download.windowsupdate.com
1 ctldl.windowsupdate.com

DNS Queries:

Observed Filenames/URLs:
3 /c/msdownload/update/software/defu/2022/01/am_delta_patch.1.355.1569.0_f5fe52e10a18f6ce4bc39541e4aae4a97c270c81.exe
2 spoonwatch.net\Policies\{31B2F340-016D-11D2-945F-00C04FB984F9}\gpt.ini
1 spoonwatch.net\Policies\{31B2F340-016D-11D2-945F-00C04FB984F9}\Machine\Registry.pol
1 spoonwatch.net\Policies\{31B2F340-016D-11D2-945F-00C04FB984F9}\Machine\Microsoft\Windows NT\SecEdit\GptTmpl.inf
1 /osk//6.jpg
1 /osk//2.jpg
1 /osk//1.jpg
1 /osk/
1 /c/msdownload/update/others/2022/01/35969632_c422aaaf8becdd90e50fb6936708c77c7f97631f.cab
1 /c/msdownload/update/others/2022/01/35969516_5bcacf676a3e98426394e538836033ea5ce2bb8c3.cab
1 /c/msdownload/update/others/2022/01/35969515_2975e5b79f9857b7a2d7aa07e0d43359735c8e80.cab

```

When did the attack begin? When did it end? Approximately how long did it take?

To determine the time of the attack, an analysis of the timestamps of events in the log was performed. The start of network activity, i.e., the first event in the log, and the end of activity, i.e., the last event in the log, were determined. After sorting the timestamps, the approximate duration of the attack was determined by calculating the difference between the first and last events.

```

#!/bin/bash
log_file="eve.json"
begin_time=$(jq -r 'select(.event_type="alert") | .timestamp' "$log_file" | sort | head -n 1)
finish_time=$(jq -r 'select(.event_type="alert") | .timestamp' "$log_file" | sort | tail -n 1)
begin_unix=$(date -d "$begin_time" +%s)
finish_unix=$(date -d "$finish_time" +%s)
duration_sec=$((finish_unix - begin_unix))
echo "Begin time: $begin_time"
echo "End time: $finish_time"
echo "Duration: $duration_sec seconds"
echo ""
echo "IP Addresses Involved (Alerts)"
jq -r 'select(.event_type="alert") | "Src: \(.src_ip) -- Dest: \(.dest_ip)"' "$log_file" | sort | uniq -c | sort -nr

```

Result:

```

(kali@kali)-[~/suricata_lab/lab1]
$ ./suricata_sum.sh
Begin time: 2022-01-07T11:04:51.756669-0500
End time: 2022-01-07T11:16:39.979554-0500
Duration: 708 seconds

IP Addresses Involved (Alerts)
 10 Src: 192.168.1.216 -- Dest: 2.56.57.108
  4 Src: 2.56.57.108 -- Dest: 192.168.1.216
  4 Src: 192.168.1.216 -- Dest: 192.168.1.2
  1 Src: 192.168.1.2 -- Dest: 192.168.1.216

```

What is/are the IP address/addresses of the internal/external host/hosts involved in the events?

During the analysis, unique source and destination IP addresses were identified. Internal host - 192.168.1.216. This host showed the most active interaction with external resources. External host (control server) - 2.56.57.108. Suspicious HTTP and TLS connections were established with this address.

```
#!/bin/bash
echo "Sources:"
jq -r '.src_ip' eve.json | sort -u | while read -r ip; do
    if [[ "$ip" =~ ^[0-9]{1,3}(\.[0-9]{1,3}){3}$ ]]; then
        echo "$ip"
    fi
done
echo "External Destinations:"
jq -r '.dest_ip' eve.json | sort -u | while read -r ip; do
    if [[ "$ip" =~ ^[0-9]{1,3}(\.[0-9]{1,3}){3}$ ]] && \
        [[ ! "$ip" =~ ^(192\.168\.|10\.|172\.|169\.|2[0-9]|3[01])\.[0-9]{1,3}$ ]]; then
        echo "$ip"
    fi
done
```

```
(kali@kali)~[~/suricata_lab/lab1]
$ ./suricata_sum.sh
Sources:
0.0.0.0
192.168.1.1
192.168.1.2
192.168.1.216
23.38.189.201
23.38.189.225
2.56.57.108
External Destinations:
104.212.67.47
184.50.62.43
20.189.173.1
204.79.197.200
204.79.197.203
20.50.80.210
20.54.89.15
224.0.0.251
224.0.0.252
23.38.189.201
23.38.189.225
239.255.255.250
255.255.255.255
2.56.57.108
40.83.240.146
52.168.112.67
52.178.17.2
52.179.216.235
52.183.220.149
52.249.36.204
8.253.189.126
```

What are some of the Source IDs of the rules that fire when the exploit occurs? Where are the Source IDs from?

We run Suricata with rules to extract identifiers from the log file:

```
(kali@kali)~[~/suricata_lab/lab1]
$ sudo suricata -c /etc/suricata/suricata.yaml -r 6.pcap -l ./out/ -s /etc/suricata/rules/local.rules
i: suricata: This is Suricata version 8.0.3 RELEASE running in USER mode
i: mpm-hs: Rule group caching - loaded: 113 newly cached: 0 total cacheable: 113
i: threads: Threads created -> RX: 1 W: 4 FM: 1 FR: 1 Engine started.
i: suricata: Signal Received. Stopping engine.
i: pcap: read 1 file, 5880 packets, 4628888 bytes
```

Several high-priority alerts were triggered, indicating the activity of malicious software and a potential data leak.

```
kali@kali: ~/suricata_lab/lab1
$ cat out

kali@kali: ~/suricata_lab/lab1/out
$ ls
eve.json  fast.log  stats.log  suricata.log

kali@kali: ~/suricata_lab/lab1/out
$ cat fast.log
01/07/2022-11:04:51.756669 [[*]] [1:2067085:1] ET INFO NTLM Session Setup Request - Negotiate [[*]] [Classification: Misc activity] [Priority: 3] [TCP] 192.168.1.216:49716 → 192.168.1.2:445
01/07/2022-11:04:51.757182 [[*]] [1:2067086:1] ET INFO NTLMv1 Session Setup Response - Challenge [[*]] [Classification: Misc activity] [Priority: 3] [TCP] 192.168.1.2:445 → 192.168.1.216:49716
01/07/2022-11:04:51.757908 [[*]] [1:2067087:1] ET INFO NTLM Session Setup Request - Auth [[*]] [Classification: Misc activity] [Priority: 2] [TCP] 192.168.1.216:49716 → 192.168.1.2:445
01/07/2022-11:07:32.344498 [[*]] [1:2034813:5] ET MALWARE Vidar/Arkei/Megumin/Oski Stealer HTTP POST Pattern [[*]] [Classification: A Network Trojan was detected] [Priority: 1] [TCP] 192.168.1.216:49738 → 2.56.57.108:80
7.108:80
01/07/2022-11:07:32.605261 [[*]] [1:2018959:6] ET INFO PE EXE or DLL Windows file download HTTP [[*]] [Classification: Potential Corporate Privacy Violation] [Priority: 1] [TCP] 2.56.57.108:80 → 192.168.1.216:49738
7.108:80
01/07/2022-11:07:32.685261 [[*]] [1:2021076:3] ET HUNTING SUSPICIOUS Dotted Quad Host M2 Response [[*]] [Classification: Potentially Bad Traffic] [Priority: 2] [TCP] 2.56.57.108:80 → 192.168.1.216:49738
01/07/2022-11:07:32.881922 [[*]] [1:2034813:5] ET MALWARE Vidar/Arkei/Megumin/Oski Stealer HTTP POST Pattern [[*]] [Classification: A Network Trojan was detected] [Priority: 1] [TCP] 192.168.1.216:49738 → 2.56.57.108:80
01/07/2022-11:07:32.887046 [[*]] [1:2021076:3] ET HUNTING SUSPICIOUS Dotted Quad Host M2 Response [[*]] [Classification: Potentially Bad Traffic] [Priority: 2] [TCP] 2.56.57.108:80 → 192.168.1.216:49738
7.108:80
01/07/2022-11:07:33.584895 [[*]] [1:2034813:5] ET MALWARE Vidar/Arkei/Megumin/Oski Stealer HTTP POST Pattern [[*]] [Classification: A Network Trojan was detected] [Priority: 1] [TCP] 192.168.1.216:49738 → 2.56.57.108:80
01/07/2022-11:07:33.589437 [[*]] [1:2021076:3] ET HUNTING SUSPICIOUS Dotted Quad Host M2 Response [[*]] [Classification: Potentially Bad Traffic] [Priority: 2] [TCP] 2.56.57.108:80 → 192.168.1.216:49738
7.108:80
01/07/2022-11:07:33.990905 [[*]] [1:2034813:5] ET MALWARE Vidar/Arkei/Megumin/Oski Stealer HTTP POST Pattern [[*]] [Classification: A Network Trojan was detected] [Priority: 1] [TCP] 192.168.1.216:49738 → 2.56.57.108:80
7.108:80
01/07/2022-11:07:34.079443 [[*]] [1:2034813:5] ET MALWARE Vidar/Arkei/Megumin/Oski Stealer HTTP POST Pattern [[*]] [Classification: A Network Trojan was detected] [Priority: 1] [TCP] 192.168.1.216:49738 → 2.56.57.108:80
7.108:80
01/07/2022-11:16:39.979554 [[*]] [1:2260002:1] SURICATA Applayer Detect protocol only one direction [[*]] [Classification: Generic Protocol Command Decode] [Priority: 3] [TCP] 192.168.1.216:49763 → 192.168.1.2:139
01/07/2022-11:07:34.379974 [[*]] [1:2034813:5] ET MALWARE Vidar/Arkei/Megumin/Oski Stealer HTTP POST Pattern [[*]] [Classification: A Network Trojan was detected] [Priority: 1] [TCP] 192.168.1.216:49738 → 2.56.57.108:80
01/07/2022-11:07:34.894408 [[*]] [1:2034813:5] ET MALWARE Vidar/Arkei/Megumin/Oski Stealer HTTP POST Pattern [[*]] [Classification: A Network Trojan was detected] [Priority: 1] [TCP] 192.168.1.216:49738 → 2.56.57.108:80
01/07/2022-11:07:35.790718 [[*]] [1:2034813:5] ET MALWARE Vidar/Arkei/Megumin/Oski Stealer HTTP POST Pattern [[*]] [Classification: A Network Trojan was detected] [Priority: 1] [TCP] 192.168.1.216:49738 → 2.56.57.108:80
01/07/2022-11:07:35.954561 [[*]] [1:2034813:5] ET MALWARE Vidar/Arkei/Megumin/Oski Stealer HTTP POST Pattern [[*]] [Classification: A Network Trojan was detected] [Priority: 1] [TCP] 192.168.1.216:49738 → 2.56.57.108:80
01/07/2022-11:07:35.956770 [[*]] [1:2033886:1] ET HUNTING Suspicious Zipped Filename in Outbound POST Request (Chrome.Default.txt) [[*]] [Classification: Potentially Bad Traffic] [Priority: 2] [TCP] 192.168.1.216:49738 → 2.56.57.108:80
01/07/2022-11:16:09.962018 [[*]] [1:2260002:1] SURICATA Applayer Detect protocol only one direction [[*]] [Classification: Generic Protocol Command Decode] [Priority: 3] [TCP] 192.168.1.216:49761 → 192.168.1.2:139
```

The most critical is SID 2034813 (triggered 9 times), which signals data leakage via Oski Stealer. SIDs 2018959, 2021076, and 2033886 (related to the download and operation of malicious software) were also triggered.

Where are the Source IDs from?

```
kali@kali: ~/suricata_lab/lab1
$ grep -o "ET MALWARE Vidar/Arkei/Megumin/Oski Stealer HTTP POST Pattern" /var/lib/suricata/rules/suricata.rules
#3987 alert http $HOME_NET any to $EXTERNAL_NET any (msg:"ET MALWARE Vidar/Arkei/Megumin/Oski Stealer HTTP POST Pattern"; flow:established,to_server; http.method:content:"POST"; http.header:content:"boundary=1BEF0A57BE110FD467A"; fast_pattern; reference:md5,dd5e5142ba2ab5f31e5518396c45baf; classtype:trojan-activity; sid:2034813; rev:5; metadata:affected_product Windows_XP_Vista_7_8_10_Server_32_64_Bit; attack_target Client_Endpoint; created_at 2018_08_15; deployment Perimeter; malware.family Arkei, confidence High, signature.severity Major, tag Description_Generated_By_Proofpoint_Nexus, updated_at 2020_08_28;)
#39
01/07/2022-11:07:34.379974 [[*]] [1:2034813:5] ET MALWARE Vidar/Arkei/Megumin/Oski Stealer HTTP POST Pattern [[*]] [Classification: A Network Trojan was detected] [Priority: 1] [TCP] 192.168.1.216:49738 → 2.56.57.108:80
```

The prefix ET in rule names indicates that they belong to the Emerging Threats rule set. Emerging Threats is a set of rules for IDS/IPS systems (Snort and Suricata) supported by Proofpoint.

We can see what was transmitted in the photos:

The top screenshot shows a list of HTTP requests from 192.168.1.216 to 2.56.57.108, all POST requests to /osk/v6.jpg. The bottom screenshot shows the packet details for the first request, including the raw data and the decoded HTTP POST body.

Frame 1501: Packet, 539 bytes on wire (4312 bits), 539 bytes captured (4312 bits) on interface eth0, Ethernet II, Src: ASUS/TCOMPU32:58:f9 (9c:5c:8e:32:58:f9), Dst: Cisco/f6:df:0a (1c:17:d3:f:0a), Internet Protocol Version 4, Src: 192.168.1.216, Dst: 2.56.57.108, Transmission Control Protocol, Src Port: 49738, Dst Port: 80, Seq: 1, Ack: 1, Len: 485, Hypertext Transfer Protocol, MIME Multipart Media Encapsulation, Type: multipart/form-data, Boundary: "1BEF0A57BE110FD467A"

Raw data: 0000 1c 17 d3 f6 df 0a 9c 5c 8e 32 58 f9 08 00 45 00 ... 2X ... E

Decoded data: POST /osk/v6.jpg HTTP/1.1\r\n\r\nAccept: text/html, application/xml;q=0.9, application/xhtml+xml, image/png, image/jpeg, image/gif, image/x-xpml\r\nAccept-Charset: iso-8859-1, utf-8, utf-16, *q=0.1\r\nAccept-Encoding: deflate, gzip, x-gzip, identity, *q=0\r\nContent-Type: multipart/form-data; boundary=1BEF0A57BE110FD467A\r\nContent-Length: 25\r\nHost: 2.56.57.108\r\nConnection: Keep-Alive\r\nCache-Control: no-cache\r\n\r\n[Response in frame: 1639]\r\n[Full request URI: http://2.56.57.108/osk/v6.jpg]\r\nFile Data: 25 bytes\r\nMIME Multipart Media Encapsulation, Type: multipart/form-data, Boundary: "1BEF0A57BE110FD467A"

What is the operating system running on the internal/external jost?

```
(kali㉿kali)-[~/suricata_lab/lab1]
$ sudo suricata -r 6.pcap -l ./out
[sudo] password for kali:
i: suricata: This is Suricata version 8.0.3 RELEASE running in USER mode
i: mpm-hs: Rule group caching - loaded: 113 newly cached: 0 total cacheable: 113
i: threads: Threads created → RX: 1 W: 4 FM: 1 FR: 1 Engine started.
i: suricata: Signal Received. Stopping engine.
i: pcap: read 1 file, 5880 packets, 462888 bytes
```

```
(kali㉿kali)-[~/suricata_lab/lab1/out]
$ jq 'select(.event_type=="alert") | {time: .timestamp, src: .src_ip, dst: .dest_ip, product: .alert.metadata.affected_product}' eve.json
{
  "time": "2022-01-07T11:04:51.756669-0500",
  "src": "192.168.1.216",
  "dst": "192.168.1.2",
  "product": [
    "Windows_XP_Vista_7_8_10_Server_32_64_Bit"
  ]
}

{
  "time": "2022-01-07T11:04:51.757182-0500",
  "src": "192.168.1.2",
  "dst": "192.168.1.216",
  "product": [
    "Windows_XP_Vista_7_8_10_Server_32_64_Bit"
  ]
}

{
  "time": "2022-01-07T11:04:51.757908-0500",
  "src": "192.168.1.216",
  "dst": "192.168.1.2",
  "product": [
    "Windows_XP_Vista_7_8_10_Server_32_64_Bit"
  ]
}

{
  "time": "2022-01-07T11:07:32.344498-0500",
  "src": "192.168.1.216",
  "dst": "2.56.57.108",
  "product": [
    "Windows_XP_Vista_7_8_10_Server_32_64_Bit"
  ]
}

{
  "time": "2022-01-07T11:07:32.605261-0500",
  "src": "2.56.57.108",
  "dst": "192.168.1.216",
  "product": null
}

{
  "time": "2022-01-07T11:07:32.605261-0500",
  "src": "2.56.57.108",
  "dst": "192.168.1.216",
  "product": null
}
```

```
(kali㉿kali)-[~/suricata_lab/lab1/out]
$ jq -e 'map(select(.event_type=="alert")) | sort_by(.timestamp) | .[] | {timestamp, signature_id: .alert.signature_id, signature: .alert.signature}' eve.json
{
  "timestamp": "2022-01-07T11:04:51.756669-0500",
  "signature_id": 2067085,
  "signature": "ET INFO NTLM Session Setup Request - Negotiate"
}

{
  "timestamp": "2022-01-07T11:04:51.756669-0500",
  "signature_id": 2067085,
  "signature": "ET INFO NTLM Session Setup Request - Negotiate"
}

{
  "timestamp": "2022-01-07T11:04:51.757182-0500",
  "signature_id": 2067086,
  "signature": "ET INFO NTLMv1 Session Setup Response - Challenge"
}

{
  "timestamp": "2022-01-07T11:04:51.757182-0500",
  "signature_id": 2067086,
  "signature": "ET INFO NTLMv1 Session Setup Response - Challenge"
}

{
  "timestamp": "2022-01-07T11:04:51.757908-0500",
  "signature_id": 2067087,
  "signature": "ET INFO NTLM Session Setup Request - Auth"
}

{
  "timestamp": "2022-01-07T11:04:51.757908-0500",
  "signature_id": 2067087,
  "signature": "ET INFO NTLM Session Setup Request - Auth"
}

{
  "timestamp": "2022-01-07T11:07:32.344498-0500",
  "signature_id": 2034813,
  "signature": "ET MALWARE Vidar/Arkei/Megumin/Oski Stealer HTTP POST Pattern"
}

{
  "timestamp": "2022-01-07T11:07:32.344498-0500",
  "signature_id": 2034813,
  "signature": "ET MALWARE Vidar/Arkei/Megumin/Oski Stealer HTTP POST Pattern"
}

{
  "timestamp": "2022-01-07T11:07:32.605261-0500",
  "signature_id": 2018959,
  "signature": null
}
```

```
{
  "timestamp": "2022-01-07T11:07:32.344498-0500",
  "signature_id": 2034813,
  "signature": "ET MALWARE Vidar/Arkei/Megumin/Oski Stealer HTTP POST Pattern"
}

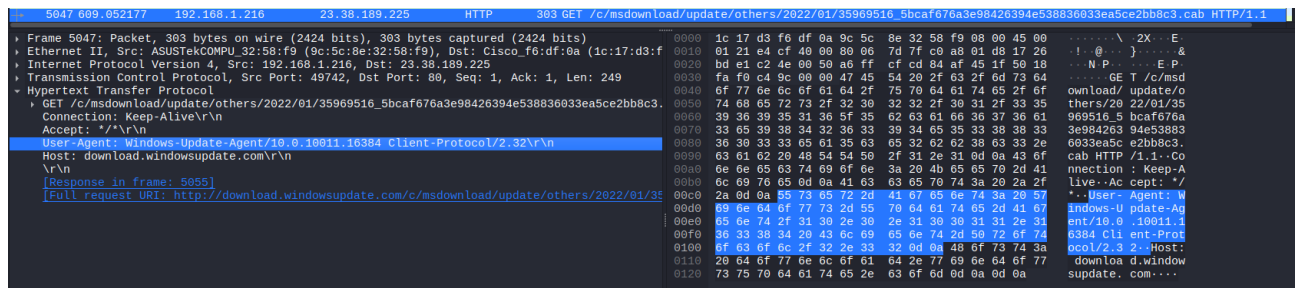
{
  "timestamp": "2022-01-07T11:07:32.605261-0500",
  "signature_id": 2018959,
  "signature": "ET INFO PE EXE or DLL Windows file download HTTP"
}

{
  "timestamp": "2022-01-07T11:07:32.605261-0500",
  "signature_id": 2021076,
  "signature": "ET HUNTING SUSPICIOUS Dotted Quad Host MZ Response"
}

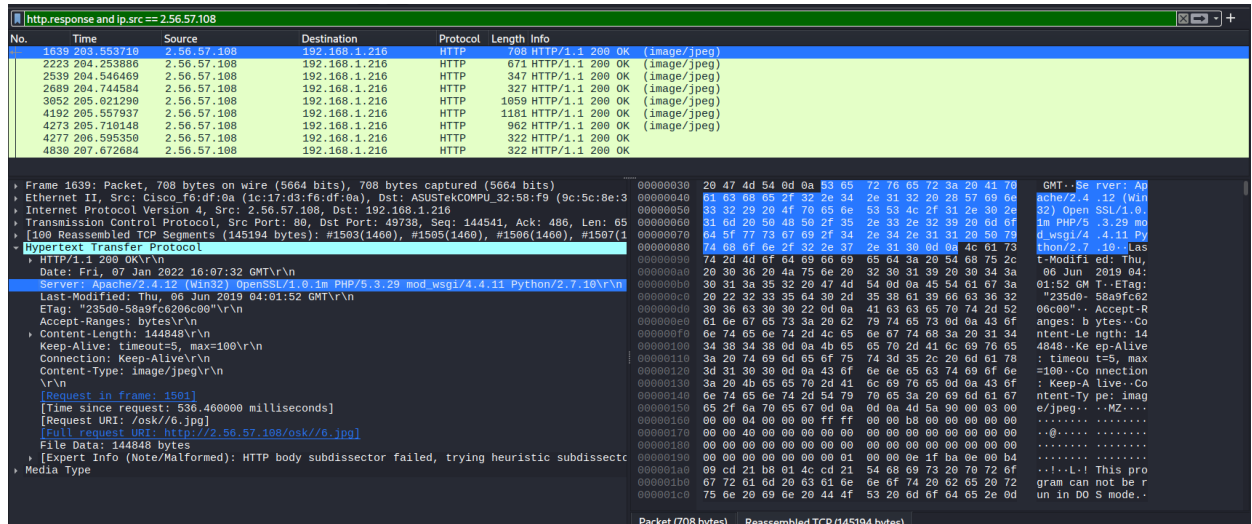
{
  "timestamp": "2022-01-07T11:07:32.605261-0500",
  "signature_id": 2018959,
  "signature": "ET INFO PE EXE or DLL Windows file download HTTP"
}

{
  "timestamp": "2022-01-07T11:07:32.605261-0500",
  "signature_id": 2021076,
  "signature": "ET HUNTING SUSPICIOUS Dotted Quad Host MZ Response"
}
```

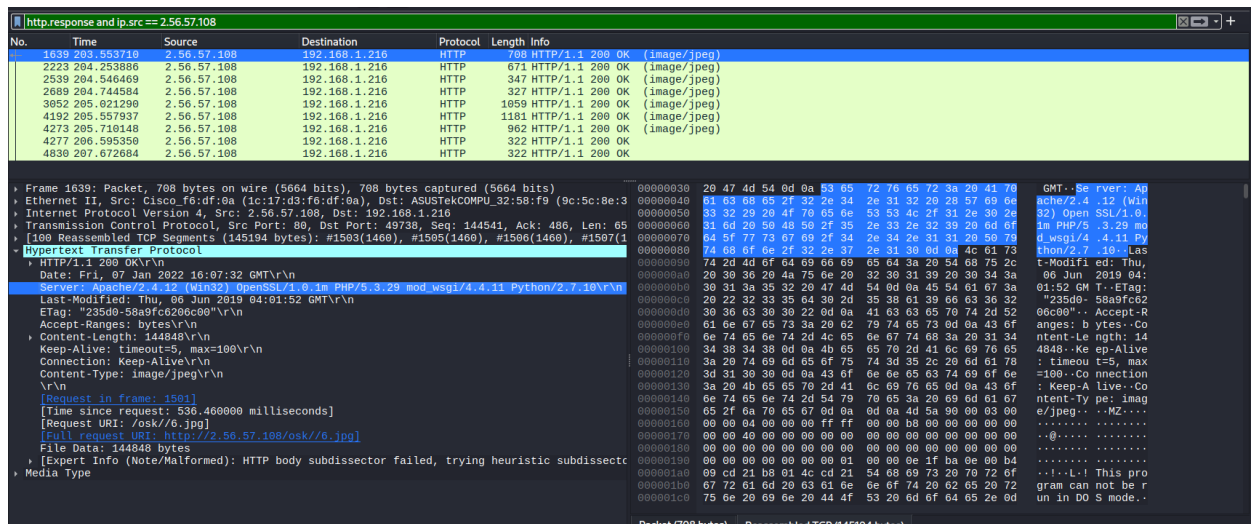
We can open the file in Wireshark and check the User-Agent:



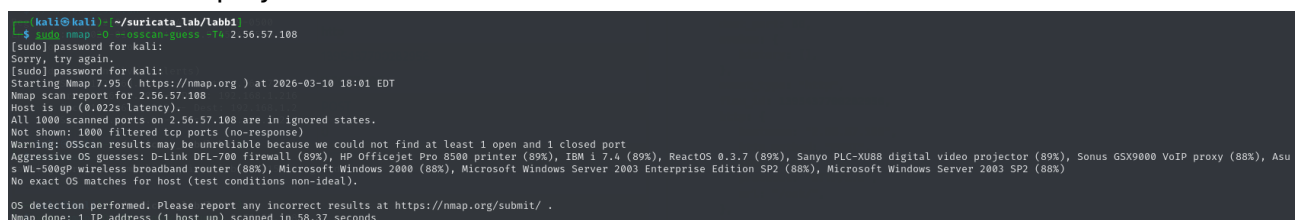
Internal host: Most likely operating system - Microsoft Windows 10. This is confirmed by traffic analysis in Wireshark and also by the metadata of the triggered Suricata rules.



The network packet has a Time to Live (TTL) value of 128, which is standard for Windows.



The HTTP response contains the header Server: Apache/2.4.12 (Win32), which indicates that the server is deployed on Windows architecture.



External host: The management server runs on Windows. The most likely systems are Microsoft Windows Server 2003 and Windows 2000.

What is the domain name that delivered the payload?

```
kali@kali: ~/suricata_lab/lab1/out
$ cat fast.log | grep -E "Vidar/Arkei" | head -n 1
01/07/2022-11:07:32.344498 [**] [1:2034813:5] ET MALWARE Vidar/Arkei/Megumin/Oski Stealer HTTP POST Pattern [**] [Classification: A Network Trojan was detected] [Priority: 1] [TCP] 192.168.1.216:49738 → 2.56.57.105:80
```

During the analysis, it was determined that the malicious software belongs to the family: Vidar/Arkei/Oski Stealer

The malicious payload was delivered without using a domain name. In the intercepted HTTP requests to the control server, the hostname field contains only the attacker's IP address. During the analysis of DNS and SMB traffic, the domain spoonwatch.net was detected, but it is likely to be an internal local domain of the victim.

Top DNS Requests	
#	Hostname
8	v10.events.data.microsoft.com
8	wpad.localdomain
8	wpad.spoonwatch.net
6	_ldap._tcp.Default-First-Site-Name._sites.dc._msdcs.spoonwatch.net
6	www.bing.com
4	settings-win.data.microsoft.com
4	_ldap._tcp.dc._msdcs.localdomain.spoonwatch.net
2	fe3cr.delivery.mp.microsoft.com
2	_ldap._tcp.Default-First-Site-Name._sites.DomainDnsZones.spoonwatch.net
2	fe2cr.update.microsoft.com

Analysis of the Top DNS Requests table shows only system requests (Microsoft, Bing) and requests to the local corporate domain spoonwatch.net.

Conclusion: During the lab work, traffic analysis was performed using the Suricata intrusion detection system and the Wireshark tool. The total number of events recorded during the attack was determined, and IDS rules were triggered, indicating the activity of malicious software. An internal infected host was identified, an external control server was determined, and network activity between them was analyzed. Signs of malicious software from the Vidar/Arkei/Oski Stealer family were also identified.